

Resumes: Courage — Answers

Courage

The abstract data generating mechanism coming out of Justice is:

$$\text{logit}(\Pr(Y = 1)) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$$

Candidate Models

Create a candidate model, modeling the outcome `call` as a function of the covariate `city`. Call it `fit_city`.

```
fit_city <- logistic_reg() |>
  fit(call ~ city, data = resumes)

fit_city |>
  tidy(conf.int = TRUE) |>
  select(term, estimate, conf.low, conf.high) |>
  mutate(across(where(is.numeric), \(x) round(x, 2)))
```

```
# A tibble: 2 x 4
  term      estimate conf.low conf.high
<chr>      <dbl>    <dbl>    <dbl>
1 (Intercept) -2.23    -2.38    -2.09
2 citychicago -0.4     -0.6     -0.19
```

The coefficient on `citychicago` is about -0.40 with a 95% confidence interval excluding zero: callbacks are less likely in Chicago than in Boston. Location matters, but it says nothing yet about the question both scenarios actually care about — so we add the name's ethnicity next.

Create a candidate model, modeling `call` as a function of both `city` and `ethnicity`. Call it `fit_city_ethnicity`.

```
fit_city_ethnicity <- logistic_reg() |>
  fit(call ~ city + ethnicity, data = resumes)

fit_city_ethnicity |>
  tidy(conf.int = TRUE) |>
  select(term, estimate, conf.low, conf.high) |>
  mutate(across(where(is.numeric), \(x) round(x, 2)))
```

```
# A tibble: 3 x 4
  term          estimate conf.low conf.high
<chr>          <dbl>    <dbl>    <dbl>
1 (Intercept)   -2.47    -2.66    -2.29
2 citychicago  -0.4     -0.61   -0.19
3 ethnicitycauc  0.44     0.23     0.65
```

The coefficient on `ethnicitycauc` is about 0.44 with a confidence interval excluding zero — a white-sounding name raises the log-odds of a callback, and the city coefficient barely moves, which is what randomized name assignment should produce. The EDA suggested resume quality might matter differently by ethnicity, so we test that next.

Create a candidate model which adds `quality`, and its interaction with `ethnicity`, to `fit_city_ethnicity`. Call it `fit_quality`.

```
fit_quality <- logistic_reg() |>
  fit(call ~ city + ethnicity + quality + ethnicity * quality, data = resumes)

fit_quality |>
  tidy(conf.int = TRUE) |>
  select(term, estimate, conf.low, conf.high) |>
  mutate(across(where(is.numeric), \(x) round(x, 2)))
```

```
# A tibble: 5 x 4
  term          estimate conf.low conf.high
<chr>          <dbl>    <dbl>    <dbl>
1 (Intercept)   -2.43    -2.68    -2.19
2 citychicago  -0.4     -0.61   -0.19
3 ethnicitycauc  0.52     0.24     0.81
4 qualitylow    -0.09    -0.41     0.24
5 ethnicitycauc:qualitylow -0.18    -0.6     0.24
```

Both the `quality` main effect and its interaction with `ethnicity` have confidence intervals that comfortably include zero. The visual pattern from the EDA does not survive this simple model, so we drop `quality` and instead keep covariates whose intervals exclude zero — plus `sex`, which the historian’s question forces us to include.

The Data Generating Mechanism

Which model should we choose, and why? Define `fit_resumes` as your chosen model and write out the fitted equation.

```
fit_resumes <- logistic_reg() |>
  fit(call ~ sex + city + ethnicity + special, data = resumes)

fit_resumes |>
  tidy(conf.int = TRUE) |>
  select(term, estimate, conf.low, conf.high) |>
  mutate(across(where(is.numeric), \(x) round(x, 2)))
```

```
# A tibble: 5 x 4
  term          estimate conf.low conf.high
  <chr>          <dbl>    <dbl>    <dbl>
1 (Intercept)   -2.75    -2.98    -2.52
2 sexmale       -0.13    -0.39     0.13
3 citychicago  -0.46    -0.68    -0.25
4 ethnicitycauc  0.45     0.24     0.66
5 specialyes     0.82     0.61     1.03
```

We choose `call ~ sex + city + ethnicity + special`. The `city`, `ethnicity`, and `special` (special skills) coefficients all have confidence intervals excluding zero. The `sex` coefficient’s interval includes zero, but the historian’s question explicitly asks about differences by sex, so the question — not the data — forces it into the model. `Quality` stays out, per `fit_quality` above.

$$\text{logit}(\Pr(\text{call} = \text{yes})) = -2.75 - 0.13 \cdot \text{male} - 0.46 \cdot \text{chicago} + 0.45 \cdot \text{caucasian} + 0.82 \cdot \text{special}$$

Each indicator is 0/1. The intercept describes a female applicant in Boston with a black-sounding name and no special skills; each coefficient shifts the log-odds of a callback. **This is our data generating mechanism.**

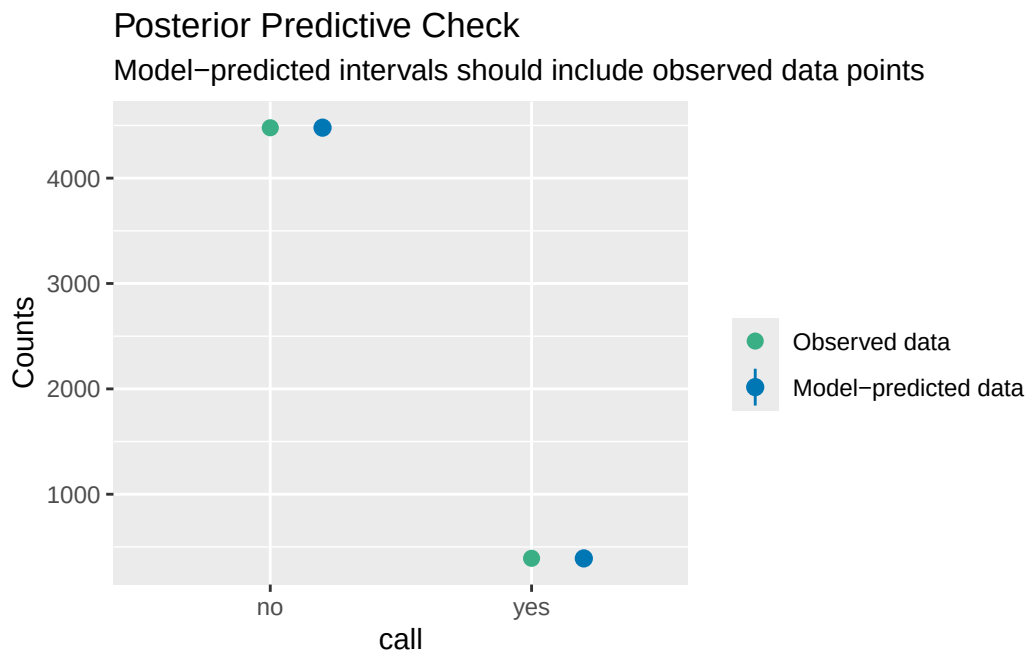
Model Checking

Use `check_predictions(extract_fit_engine(fit_resumes))` to perform a posterior predictive check of your chosen model.

```
check_predictions(extract_fit_engine(fit_resumes))
```

Ignoring unknown labels:

```
* size : ""  
* alpha : ""
```



`check_predictions()` simulates new callback data from the fitted model and compares it to the data we observed. The simulated replicates bracket the observed share of callbacks, so the model is not obviously misdescribing the outcome — the check we want to pass before trusting the predictions in Temperance.